

Half-Car Lumped-Parameter Model

Assumptions

• Small angles: disturbance amplitude will be small compared to car body.

• Rigid Bodies simplify calculations for ride comfort

• Bogie Pitch ignored track disturbances imparting ride comfort are at lower frequencies compared to what pitches bogie frame

• Front & Rear bogie wheelsets combined: since pitching is ignored



Car-Body

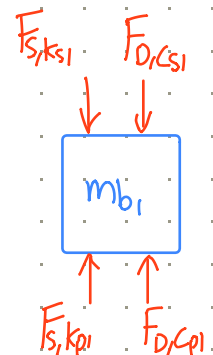
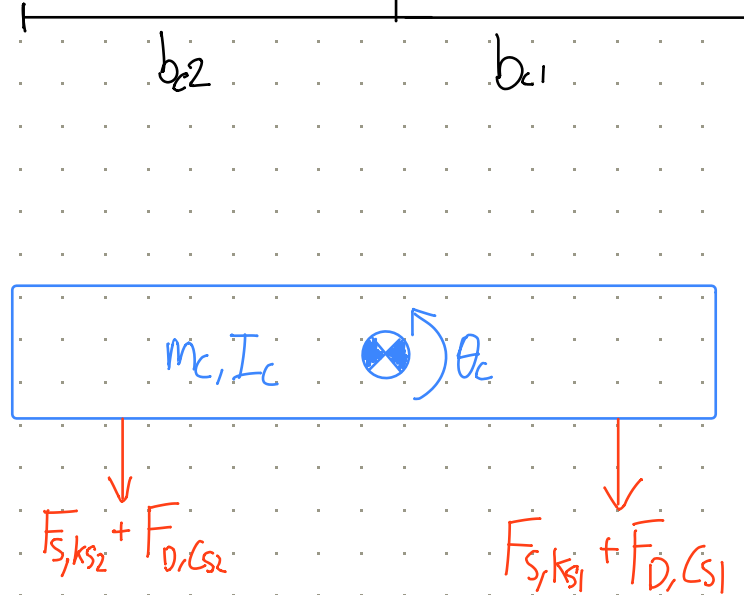
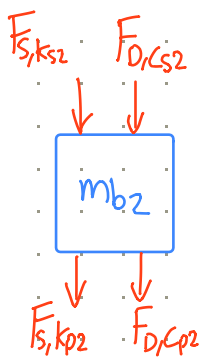
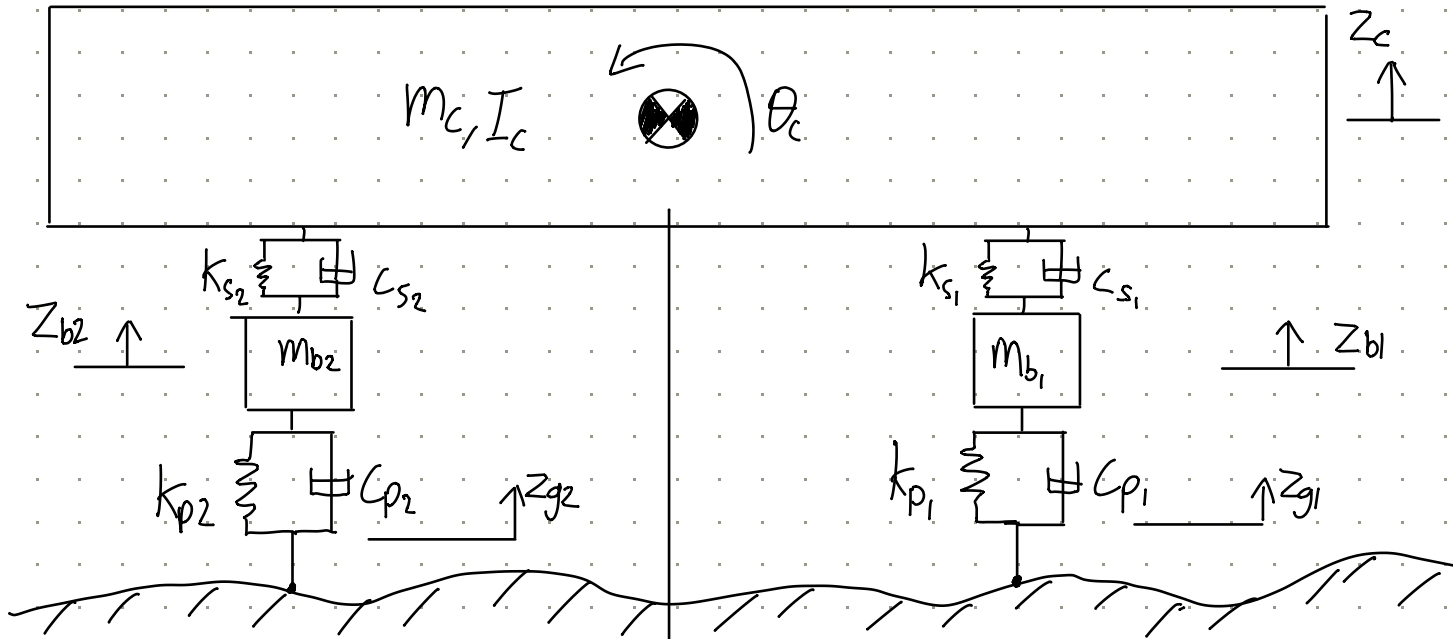
Secondary-Suspension

Bogies

Primary-Suspension

Wheels

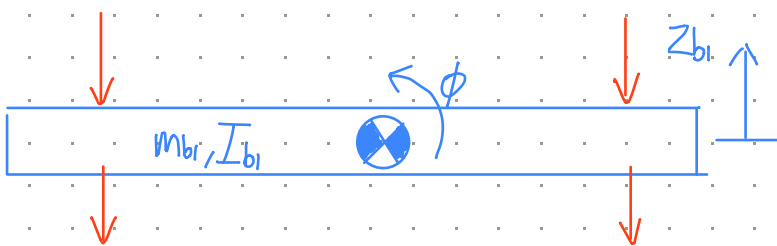
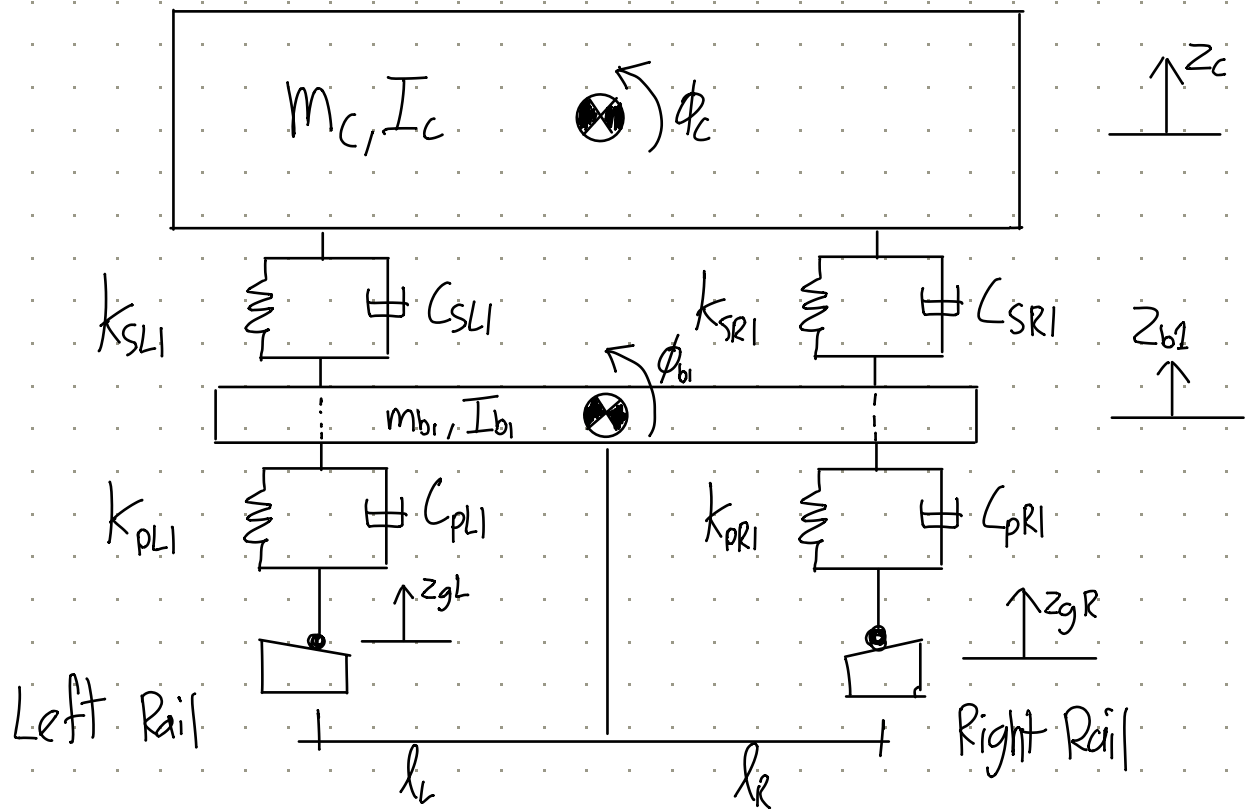
Rail



$$\sum T = I_c \ddot{\theta}_c = -b_{c2}(F_{s,k_s2} + F_{D,c_s2}) + b_{c1}(F_{s,k_s1} + F_{D,c_s1})$$

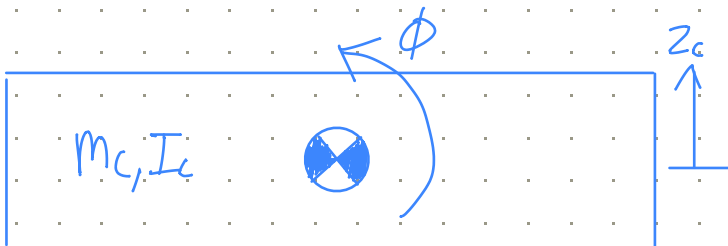
$$z_c = \frac{(F_{s,k_s2} + F_{D,c_s2}) + (F_{s,k_s1} + F_{D,c_s1})}{2}$$

Lateral View (Front Half)



$$I_{b1} \ddot{\phi}_{b1} = l_L [(F_{S,k_{PL1}} + F_{D,C_{PL1}}) + (F_{S,k_{SL1}} + F_{D,C_{SL1}})] - l_R [(F_{S,k_{PR1}} + F_{D,C_{PR1}}) + (F_{S,k_{SR1}} + F_{D,C_{SR1}})]$$

$$z_{b1} = \frac{(F_{S,k_{PL1}} + F_{D,C_{PL1}}) + (F_{S,k_{SL1}} + F_{D,C_{SL1}}) + (F_{S,k_{PR1}} + F_{D,C_{PR1}}) + (F_{S,k_{SR1}} + F_{D,C_{SR1}})}{2}$$



$$I_c \ddot{\phi}_c = l_L (F_{S,k_{SL1}} + F_{D,C_{SL1}}) + l_R (F_{S,k_{SR1}} + F_{D,C_{SR1}})$$

$$z_c = \frac{(F_{S,k_{SL1}} + F_{D,C_{SL1}}) + (F_{S,k_{SR1}} + F_{D,C_{SR1}})}{2}$$

Final 7 Equations of Motion

Bogie Bounce 1 $m_{b1} \ddot{z}_{b1} =$

$$\begin{aligned} & -k_{PL1}[(z_{b1} - l_L \phi_{b1}) - z_{g1}] - C_{PL1}[(\dot{z}_{b1} - l_L \dot{\phi}_{b1}) - \dot{z}_{g1}] - k_{SL1}[(z_{b1} - l_L \phi_{b1}) - (z_c - l_L \phi_c + b_{c1} \theta_c)] \\ & - C_{SL1}[(\dot{z}_{b1} - l_L \dot{\phi}_{b1}) - (\dot{z}_c - l_L \dot{\phi}_c + b_{c1} \dot{\theta}_c)] - k_{PR1}[(z_{b1} + l_R \phi_{b1}) - z_{gR}] - C_{PR1}[(\dot{z}_{b1} + l_R \dot{\phi}_{b1}) - \dot{z}_{gR}] \\ & - k_{SR1}[(z_{b1} + l_R \phi_{b1}) - (z_c + l_R \phi_c + b_{c1} \theta_c)] - C_{SR1}[(\dot{z}_{b1} + l_R \dot{\phi}_{b1}) - (\dot{z}_c + l_R \dot{\phi}_c + b_{c1} \dot{\theta}_c)] \end{aligned}$$

Bogie Roll 1 $I_{b1} \ddot{\phi}_{b1} =$

$$\begin{aligned} & -l_L \left[-k_{PL1}[(z_{b1} - l_L \phi_{b1}) - z_{g1}] - C_{PL1}[(\dot{z}_{b1} - l_L \dot{\phi}_{b1}) - \dot{z}_{g1}] - k_{SL1}[(z_{b1} - l_L \phi_{b1}) - (z_c - l_L \phi_c + b_{c1} \theta_c)] \right. \\ & \left. - C_{SL1}[(\dot{z}_{b1} - l_L \dot{\phi}_{b1}) - (\dot{z}_c - l_L \dot{\phi}_c + b_{c1} \dot{\theta}_c)] \right] + l_R \left[-k_{PR1}[(z_{b1} + l_R \phi_{b1}) - z_{gR}] - C_{PR1}[(\dot{z}_{b1} + l_R \dot{\phi}_{b1}) - \dot{z}_{gR}] \right. \\ & \left. - k_{SR1}[(z_{b1} + l_R \phi_{b1}) - (z_c + l_R \phi_c + b_{c1} \theta_c)] - C_{SR1}[(\dot{z}_{b1} + l_R \dot{\phi}_{b1}) - (\dot{z}_c + l_R \dot{\phi}_c + b_{c1} \dot{\theta}_c)] \right] \end{aligned}$$

Bogie Bounce 2 $m_{b2} \ddot{z}_{b2} =$

$$\begin{aligned} & -k_{PL2}[(z_{b2} - l_L \phi_{b2}) - z_{g1}] - C_{PL2}[(\dot{z}_{b2} - l_L \dot{\phi}_{b2}) - \dot{z}_{g1}] - k_{SL2}[(z_{b2} - l_L \phi_{b2}) - (z_c - l_L \phi_c - b_{c2} \theta_c)] \\ & - C_{SL2}[(\dot{z}_{b2} - l_L \dot{\phi}_{b2}) - (\dot{z}_c - l_L \dot{\phi}_c - b_{c2} \dot{\theta}_c)] - k_{PR2}[(z_{b2} + l_R \phi_{b2}) - z_{gR}] - C_{PR2}[(\dot{z}_{b2} + l_R \dot{\phi}_{b2}) - \dot{z}_{gR}] \\ & - k_{SR2}[(z_{b2} + l_R \phi_{b2}) - (z_c + l_R \phi_c - b_{c2} \theta_c)] - C_{SR2}[(\dot{z}_{b2} + l_R \dot{\phi}_{b2}) - (\dot{z}_c + l_R \dot{\phi}_c - b_{c2} \dot{\theta}_c)] \end{aligned}$$

Bogie Roll 2 $I_{b2} \ddot{\phi}_{b2} =$

$$\begin{aligned} & -l_L \left[-k_{PL2}[(z_{b2} - l_L \phi_{b2}) - z_{g1}] - C_{PL2}[(\dot{z}_{b2} - l_L \dot{\phi}_{b2}) - \dot{z}_{g1}] - k_{SL2}[(z_{b2} - l_L \phi_{b2}) - (z_c - l_L \phi_c - b_{c2} \theta_c)] \right. \\ & \left. - C_{SL2}[(\dot{z}_{b2} - l_L \dot{\phi}_{b2}) - (\dot{z}_c - l_L \dot{\phi}_c - b_{c2} \dot{\theta}_c)] \right] + l_R \left[-k_{PR2}[(z_{b2} + l_R \phi_{b2}) - z_{gR}] - C_{PR2}[(\dot{z}_{b2} + l_R \dot{\phi}_{b2}) - \dot{z}_{gR}] \right. \\ & \left. - k_{SR2}[(z_{b2} + l_R \phi_{b2}) - (z_c + l_R \phi_c - b_{c2} \theta_c)] - C_{SR2}[(\dot{z}_{b2} + l_R \dot{\phi}_{b2}) - (\dot{z}_c + l_R \dot{\phi}_c - b_{c2} \dot{\theta}_c)] \right] \end{aligned}$$

Car-Body Roll $I_{C_x} \ddot{\phi}_c =$

$$\begin{aligned} & -l_L [-k_{SL1} [(\dot{z}_c - l_L \dot{\phi}_c + b_{c1} \dot{\theta}_c) - (\dot{z}_{b1} - l_L \dot{\phi}_{b1})] - c_{SL1} [(\dot{z}_c - l_L \dot{\phi}_c + b_{c1} \dot{\theta}_c) - (\dot{z}_{b1} - l_L \dot{\phi}_{b1})] - k_{SL2} [(\dot{z}_c - l_L \dot{\phi}_c - b_{c2} \dot{\theta}_c) - (\dot{z}_{b2} - l_L \dot{\phi}_{b2})] \\ & - c_{SL2} [(\dot{z}_c - l_L \dot{\phi}_c - b_{c2} \dot{\theta}_c) - (\dot{z}_{b2} - l_L \dot{\phi}_{b2})]] + l_R [-k_{SR1} [(\dot{z}_c + l_R \dot{\phi}_c + b_{c1} \dot{\theta}_c) - (\dot{z}_{b1} + l_R \dot{\phi}_{b1})] - c_{SR1} [(\dot{z}_c + l_R \dot{\phi}_c + b_{c1} \dot{\theta}_c) - (\dot{z}_{b1} + l_R \dot{\phi}_{b1})] \\ & - k_{SR2} [(\dot{z}_c + l_R \dot{\phi}_c - b_{c2} \dot{\theta}_c) - (\dot{z}_{b1} + l_R \dot{\phi}_{b2})] - c_{SR2} [(\dot{z}_c + l_R \dot{\phi}_c - b_{c2} \dot{\theta}_c) - (\dot{z}_{b1} + l_R \dot{\phi}_{b2})]] \end{aligned}$$

Car-Body Pitch $I_{C_y} \ddot{\theta}_c =$

$$\begin{aligned} & -b_{c2} [-k_{SL2} [(\dot{z}_c - l_L \dot{\phi}_c - b_{c2} \dot{\theta}_c) - (\dot{z}_{b2} - l_L \dot{\phi}_{b2})] - c_{SL2} [(\dot{z}_c - l_L \dot{\phi}_c - b_{c2} \dot{\theta}_c) - (\dot{z}_{b2} - l_L \dot{\phi}_{b2})] - k_{SR2} [(\dot{z}_c + l_R \dot{\phi}_c - b_{c2} \dot{\theta}_c) - (\dot{z}_{b2} + l_R \dot{\phi}_{b2})] \\ & - c_{SR2} [(\dot{z}_c + l_R \dot{\phi}_c - b_{c2} \dot{\theta}_c) - (\dot{z}_{b2} + l_R \dot{\phi}_{b2})]] + b_{c1} [-k_{SL1} [(\dot{z}_c - l_L \dot{\phi}_c + b_{c1} \dot{\theta}_c) - (\dot{z}_{b1} - l_L \dot{\phi}_{b1})] \\ & - c_{SL1} [(\dot{z}_c - l_L \dot{\phi}_c + b_{c1} \dot{\theta}_c) - (\dot{z}_{b1} - l_L \dot{\phi}_{b1})] - k_{SR1} [(\dot{z}_c + l_R \dot{\phi}_c + b_{c1} \dot{\theta}_c) - (\dot{z}_{b1} + l_R \dot{\phi}_{b1})] - c_{SR1} [(\dot{z}_c + l_R \dot{\phi}_c + b_{c1} \dot{\theta}_c) - (\dot{z}_{b1} + l_R \dot{\phi}_{b1})]] \end{aligned}$$

Car-Body Bounce $m_c \ddot{z}_c =$

$$\begin{aligned} & -k_{SL1} [(\dot{z}_c - l_L \dot{\phi}_c + b_{c1} \dot{\theta}_c) - (\dot{z}_{b1} - l_L \dot{\phi}_{b1})] - c_{SL1} [(\dot{z}_c - l_L \dot{\phi}_c + b_{c1} \dot{\theta}_c) - (\dot{z}_{b1} - l_L \dot{\phi}_{b1})] - k_{SL2} [(\dot{z}_c - l_L \dot{\phi}_c - b_{c2} \dot{\theta}_c) - (\dot{z}_{b2} - l_L \dot{\phi}_{b2})] \\ & - c_{SL2} [(\dot{z}_c - l_L \dot{\phi}_c - b_{c2} \dot{\theta}_c) - (\dot{z}_{b2} - l_L \dot{\phi}_{b2})] - k_{SR1} [(\dot{z}_c + l_R \dot{\phi}_c + b_{c1} \dot{\theta}_c) - (\dot{z}_{b1} + l_R \dot{\phi}_{b1})] - c_{SR1} [(\dot{z}_c + l_R \dot{\phi}_c + b_{c1} \dot{\theta}_c) - (\dot{z}_{b1} + l_R \dot{\phi}_{b1})] \\ & - k_{SR2} [(\dot{z}_c + l_R \dot{\phi}_c - b_{c2} \dot{\theta}_c) - (\dot{z}_{b1} + l_R \dot{\phi}_{b2})] - c_{SR2} [(\dot{z}_c + l_R \dot{\phi}_c - b_{c2} \dot{\theta}_c) - (\dot{z}_{b1} + l_R \dot{\phi}_{b2})] \end{aligned}$$

M-K-C Matrices

7-DOF: $Z_c, \phi_c, \theta_c, Z_{b1}, \phi_{b1}, Z_{b2}, \phi_{b2}$

$$Z_{gL} = U_L$$

$$Z_{gR} = U_R$$

$$M = \begin{matrix} & \begin{matrix} Z_c & \phi_c & \theta_c & Z_{b1} & \phi_{b1} & Z_{b2} & \phi_{b2} \end{matrix} \\ \begin{matrix} Z_c \\ \phi_c \\ \theta_c \\ Z_{b1} \\ \phi_{b1} \\ Z_{b2} \\ \phi_{b2} \end{matrix} & \begin{bmatrix} m_c & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & I_{c_x} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I_{c_y} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & m_{b1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I_{b1} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & m_{b2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & I_{b2} \end{bmatrix} \end{matrix}$$

$$C = \begin{matrix} & \begin{matrix} \dot{Z}_c & \dot{\phi}_c & \dot{\theta}_c & \dot{Z}_{b1} & \dot{\phi}_{b1} & \dot{Z}_{b2} & \dot{\phi}_{b2} \end{matrix} \\ \begin{matrix} \dot{Z}_c \\ \dot{\phi}_c \\ \dot{\theta}_c \\ \dot{Z}_{b1} \\ \dot{\phi}_{b1} \\ \dot{Z}_{b2} \\ \dot{\phi}_{b2} \end{matrix} & \begin{bmatrix} C_{s11} + C_{s12} & -k_L(C_{s11} + C_{s12}) & (C_{s11} + C_{s12})b_{c1} & -(C_{s11} + C_{s12}) & C_{s11}k_L - C_{s12}k_R & -(C_{s12} + C_{s12}) & C_{s12}k_L - C_{s12}k_R \\ + C_{s11} + C_{s12} & +k_R(C_{s11} + C_{s12}) & (C_{s12} + C_{s12})b_{c2} & -(C_{s11} + C_{s12}) & C_{s11}k_L - C_{s12}k_R & -(C_{s12} + C_{s12}) & C_{s12}k_L - C_{s12}k_R \\ -k_L(C_{s11} + C_{s12}) & k_L(C_{s11} + C_{s12}) & k_L(C_{s11} + k_R(C_{s11})b_{c1} & k_L(C_{s11} - k_R(C_{s11}) & -k_L^2C_{s11} - k_R^2C_{s11} & k_LC_{s12} - k_R(C_{s12}) & -k_L^2C_{s12} - k_R^2C_{s12} \\ +k_R(C_{s11} + C_{s12}) & +k_R^2(C_{s11} + C_{s12}) & k_L(C_{s12} - k_R(C_{s12})b_{c2} & k_L(C_{s11} - k_R(C_{s11}) & -k_L^2C_{s11} - k_R^2C_{s11} & k_LC_{s12} - k_R(C_{s12}) & -k_L^2C_{s12} - k_R^2C_{s12} \\ (C_{s11} + C_{s11})b_{c1} & -(k_L(C_{s11} + k_R(C_{s11})b_{c1} & (C_{s11} + C_{s11})b_{c1}^2 & -(C_{s11} + C_{s11})b_{c1} & (C_{s11}k_L - C_{s12}k_R)b_{c1} & (C_{s12} + C_{s12})b_{c2} & (C_{s12}k_L - C_{s12}k_R)b_{c2} \\ -(C_{s12} + C_{s12})b_{c2} & -(k_L(C_{s12} + k_R(C_{s12})b_{c2} & + (C_{s12} + C_{s12})b_{c2}^2 & -(C_{s11} + C_{s11})b_{c1} & (C_{s11}k_L - C_{s12}k_R)b_{c1} & (C_{s12} + C_{s12})b_{c2} & (C_{s12}k_L - C_{s12}k_R)b_{c2} \\ -C_{s11} - C_{s11} & C_{s11}k_L - C_{s12}k_R & -(C_{s11} + C_{s11})b_{c1} & C_{p11} + C_{s11} & -(C_{p11} + C_{s11})k_L & 0 & 0 \\ C_{p11} + C_{s11} & C_{p11}k_L - C_{s12}k_R & -(C_{s11} + C_{s11})b_{c1} & -C_{p11} + C_{s11} & -(C_{p11} + C_{s11})k_L & 0 & 0 \\ k_L(C_{s11} - k_R(C_{s11}) & -k_L^2C_{s11} - k_R^2C_{s11} & k_L(C_{s11} - k_R(C_{s11})b_{c1} & -C_{p11} + C_{s11} & -(C_{p11} + C_{s11})k_L & 0 & 0 \\ +C_{p11} + C_{s11} & +C_{p11}k_L - C_{s12}k_R & + (C_{p11} + C_{s11})b_{c1} & +C_{p11} + C_{s11} & -(C_{p11} + C_{s11})k_L & 0 & 0 \\ -C_{s12} - C_{s12} & C_{s12}k_L - C_{s12}k_R & (C_{s12} + C_{s12})b_{c2} & 0 & 0 & C_{p12} + C_{s12} & -(C_{p12} + C_{s12})k_L \\ +C_{p12} + C_{s12} & +C_{p12}k_L - C_{s12}k_R & + (C_{p12} + C_{s12})b_{c2} & 0 & 0 & +C_{p12} + C_{s12} & -(C_{p12} + C_{s12})k_L \\ k_L(C_{s12} - k_R(C_{s12}) & -k_L^2C_{s12} - k_R^2C_{s12} & k_L(C_{s12} - k_R(C_{s12})b_{c2} & 0 & 0 & -k_L(C_{p12} + C_{s12}) & k_L^2(C_{p12} + C_{s12}) \\ +k_R(C_{p12} + C_{s12}) & +k_R^2(C_{p12} + C_{s12}) & + (k_L(C_{p12} + C_{s12})b_{c2} & 0 & 0 & +k_R(C_{p12} + C_{s12}) & +k_R^2(C_{p12} + C_{s12}) \end{bmatrix} \end{matrix}$$

Due to the coupling nature of the included springs and dampers, the K matrix will look like the C matrix but with all C's replaced with K's.

$$G_2 = \begin{array}{c|c|c|c|c} & Z_{g1} & Z_{gR1} & Z_{g12} & Z_{gR2} \\ \hline Z_c & 0 & 0 & 0 & 0 \\ \phi_c & 0 & 0 & 0 & 0 \\ \theta_c & 0 & 0 & 0 & 0 \\ \hline Z_{b1} & k_{p1} & k_{pR1} & 0 & 0 \\ \phi_{b1} & -l_L k_{p1} & l_R k_{pR1} & 0 & 0 \\ \hline Z_{b2} & 0 & 0 & k_{p12} & k_{pR2} \\ \phi_{b2} & 0 & 0 & -l_L k_{p12} & l_R k_{pR2} \end{array}$$

The following pages contain
some draft work

Matrix Prep

7-DOF: $z_c, \phi_c, \theta_c, z_{b1}, \phi_{b1}, z_{b2}, \phi_{b2}$

$$z_{gL} = u_L$$

$$z_{gR} = u_R$$

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \\ X_6 \\ X_7 \\ X_8 \\ X_9 \\ X_{10} \\ X_{11} \\ X_{12} \\ X_{13} \\ X_{14} \end{bmatrix}$$

$$\begin{cases} X_1 = z_c \\ X_2 = \phi_c \\ X_3 = \theta_c \\ X_4 = z_{b1} \\ X_5 = \phi_{b1} \\ X_6 = z_{b2} \\ X_7 = \phi_{b2} \\ X_8 = \dot{z}_c = \dot{X}_1 \\ X_9 = \dot{\phi}_c = \dot{X}_2 \\ X_{10} = \dot{\theta}_c = \dot{X}_3 \\ X_{11} = \dot{z}_{b1} = \dot{X}_4 \\ X_{12} = \dot{\phi}_{b1} = \dot{X}_5 \\ X_{13} = \dot{z}_{b2} = \dot{X}_6 \\ X_{14} = \dot{\phi}_{b2} = \dot{X}_7 \end{cases}$$

$$\textcircled{1} m_{b1} \ddot{z}_{b1} + (c_{pL1} + c_{sL1} + c_{pR1} + c_{sR1}) \dot{z}_{b1} + (k_{pL1} + k_{sL1} + k_{pR1} + k_{sR1}) z_{b1} =$$

$$(c_{pL1} + c_{sL1}) l_L \dot{\phi}_{b1} + (c_{pR1} + c_{sR1}) l_R \dot{\phi}_{b1} + (k_{pL1} + k_{sL1}) l_L \phi_{b1} + (k_{pR1} + k_{sR1}) l_R \phi_{b1}$$

$$+ (k_{pL1} + c_{pL1}) u_L + (k_{pR1} + c_{pR1}) u_R + (k_{sL1} + k_{sR1}) z_c + (c_{sL1} + c_{sR1}) \dot{z}_c$$

$$- (k_{sL1}) l_L \phi_c - (c_{sL1}) l_L \dot{\phi}_c + k_{sR1} l_R \phi_c + c_{sR1} l_R \dot{\phi}_c + (k_{sL1} + k_{sR1}) b_{c1} \theta_c + (c_{sL1} + c_{sR1}) b_{c1} \dot{\theta}_c$$

$$\textcircled{2} m_c \ddot{z}_c + (c_{sL1} + c_{sL2} + c_{sR1} + c_{sR2}) \dot{z}_c + (k_{sL1} + k_{sL2} + k_{sR1} + k_{sR2}) z_c$$

$$- [(k_{sL1} + k_{sL2}) l_L - (k_{sR1} + k_{sR2}) l_R] \phi_c - [(c_{sL1} + c_{sL2}) l_L - (c_{sR1} + c_{sR2}) l_R] \dot{\phi}_c$$

$$+ [(k_{sL1} + k_{sR1}) b_{c1} - (k_{sL2} + k_{sR2}) b_{c2}] \theta$$

$$+ [(c_{sL1} + c_{sR1}) b_{c1} - (c_{sL2} + c_{sR2}) b_{c2}] \dot{\theta} - (k_{sL1} + k_{sR1}) z_{b1} - (k_{sL2} + k_{sR2}) z_{b2}$$

$$- (c_{sL1} + c_{sR1}) \dot{z}_{b1} - (c_{sL2} + c_{sR2}) \dot{z}_{b2} + (k_{sL1} l_L - k_{sR1} l_R) \phi_{b1} + (c_{sL1} l_L - c_{sR1} l_R) \dot{\phi}_{b1}$$

$$+ (k_{sL2} l_L - k_{sR2} l_R) \phi_{b2} + (c_{sL2} l_L - c_{sR2} l_R) \dot{\phi}_{b2} = 0$$

Equations of Motion

Half-Car Equations

$$m_{b2}\ddot{z}_{b2} = -F_{s,ks2} - F_{D,Cs2} - F_{s,kp2} - F_{D,Cp2}$$

$$= -k_{s2}[z_{b2} - (z_c - b_{c2}\theta_c)] - C_{s2}[\dot{z}_{b2} - (\dot{z}_c - b_{c2}\dot{\theta}_c)] - k_{p2}(z_{b2} - z_{g2}) - C_{p2}(\dot{z}_{b2} - \dot{z}_{g2})$$

$$m_{b1}\ddot{z}_{b1} = -F_{s,ks1} - F_{D,Cs1} - F_{s,kp1} - F_{D,Cp1}$$

$$= -k_{s1}[z_{b1} - (z_c - b_{c1}\theta_c)] - C_{s1}[\dot{z}_{b1} - (\dot{z}_c + b_{c1}\dot{\theta}_c)] - k_{p1}(z_{b1} - z_{g1}) - C_{p1}(\dot{z}_{b1} - \dot{z}_{g1})$$

$$m_c\ddot{z}_c = -F_{s,ks2} - F_{D,Cs2} - F_{s,ks1} - F_{D,Cs1}$$

$$= k_{s2}[z_{b2} - (z_c - b_{c2}\theta_c)] + C_{s2}[\dot{z}_{b2} - (\dot{z}_c - b_{c2}\dot{\theta}_c)] + k_{s1}[z_{b1} - (z_c - b_{c1}\theta_c)] + C_{s1}[\dot{z}_{b1} - (\dot{z}_c + b_{c1}\dot{\theta}_c)]$$

$$I_c\ddot{\theta}_c = -b_{c2}(F_{s,ks2} - F_{D,Cs2}) + b_{c1}(F_{s,ks1} - F_{D,Cs1})$$

Lateral View Equations

$$m_{b1}\ddot{z}_{b1} = [-F_{s,kpL} - F_{D,CpL} - F_{s,ksL} - F_{D,CsL}] + [-F_{s,kpR} - F_{D,CpR} - F_{s,ksR} - F_{D,CsR}] =$$

$$-k_{pL}[z_{b1} - l_L\phi_{b1} - z_{gL}] - C_{pL}[\dot{z}_{b1} - l_L\dot{\phi}_{b1} - \dot{z}_{gL}] - k_{sL}[z_{b1} - l_L\phi_{b1} - (z_c - l_L\phi_c)] - C_{sL}[\dot{z}_{b1} - l_L\dot{\phi}_{b1} - (\dot{z}_c - l_L\dot{\phi}_c)]$$

$$-k_{pR}[z_{b1} + l_R\phi_{b1} - z_{gR}] - C_{pR}[\dot{z}_{b1} + l_R\dot{\phi}_{b1} - \dot{z}_{gR}] - k_{sR}[z_{b1} + l_R\phi_{b1} - (z_c + l_R\phi_c)] - C_{sR}[\dot{z}_{b1} + l_R\dot{\phi}_{b1} - (\dot{z}_c + l_R\dot{\phi}_c)]$$

$$I_{b1}\ddot{\phi}_{b1} = -l_L[-F_{s,kpL} - F_{D,CpL} - F_{s,ksL} - F_{D,CsL}] + l_R[-F_{s,kpR} - F_{D,CpR} - F_{s,ksR} - F_{D,CsR}] =$$

$$-l_L[-k_{pL}[z_{b1} - l_L\phi_{b1} - z_{gL}] - C_{pL}[\dot{z}_{b1} - l_L\dot{\phi}_{b1} - \dot{z}_{gL}] - k_{sL}[z_{b1} - l_L\phi_{b1} - (z_c - l_L\phi_c)] - C_{sL}[\dot{z}_{b1} - l_L\dot{\phi}_{b1} - (\dot{z}_c - l_L\dot{\phi}_c)]]$$

$$+ l_R[-k_{pR}[z_{b1} + l_R\phi_{b1} - z_{gR}] - C_{pR}[\dot{z}_{b1} + l_R\dot{\phi}_{b1} - \dot{z}_{gR}] - k_{sR}[z_{b1} + l_R\phi_{b1} - (z_c + l_R\phi_c)] - C_{sR}[\dot{z}_{b1} + l_R\dot{\phi}_{b1} - (\dot{z}_c + l_R\dot{\phi}_c)]]$$

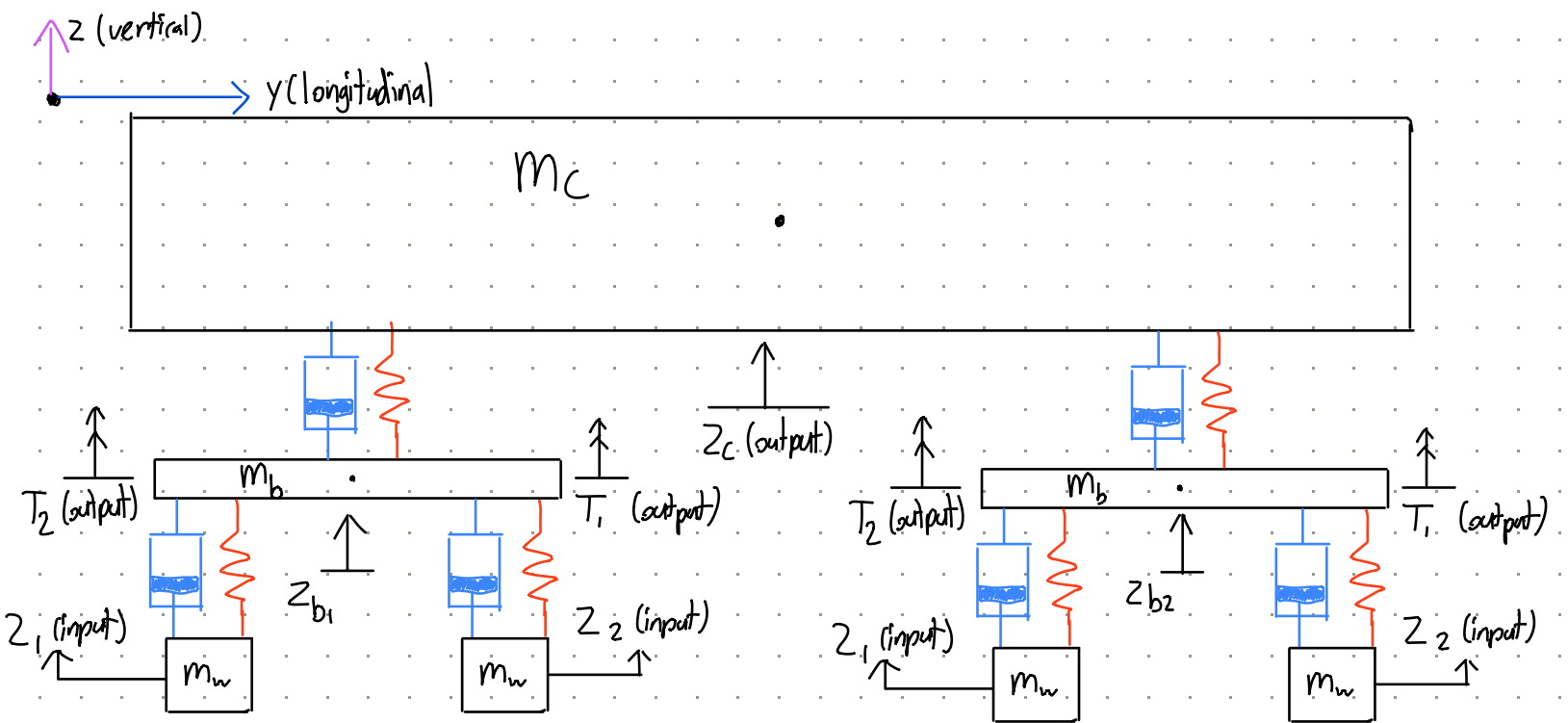
$$m_c\ddot{z}_c = -F_{s,ksL} - F_{D,CsL} - F_{s,ksR} - F_{D,CsR}$$

$$-k_{sL}[z_c - l_L\phi_c - (z_{b1} - l_L\phi_{b1})] - C_{sL}[(\dot{z}_c - l_L\dot{\phi}_c) - (\dot{z}_{b1} - l_L\dot{\phi}_{b1})] - k_{sR}[z_c + l_R\phi_c - (z_{b1} + l_R\phi_{b1})] - C_{sR}[(\dot{z}_c + l_R\dot{\phi}_c) - (\dot{z}_{b1} + l_R\dot{\phi}_{b1})]$$

$$I_c\ddot{\phi}_c = -l_L(-F_{s,ksL} - F_{D,CsL}) + l_R(-F_{s,ksR} - F_{D,CsR}) =$$

$$-l_L[-k_{sL}[z_c - l_L\phi_c - (z_{b1} - l_L\phi_{b1})] - C_{sL}[(\dot{z}_c - l_L\dot{\phi}_c) - (\dot{z}_{b1} - l_L\dot{\phi}_{b1})]] + l_R[k_{sR}[z_c + l_R\phi_c - (z_{b1} + l_R\phi_{b1})] - C_{sR}[(\dot{z}_c + l_R\dot{\phi}_c) - (\dot{z}_{b1} + l_R\dot{\phi}_{b1})]]$$

$$K = \begin{bmatrix} z_c \\ \phi_c \\ \theta_c \\ z_{b1} \\ \phi_{b1} \\ z_{b2} \\ \phi_{b2} \end{bmatrix}$$



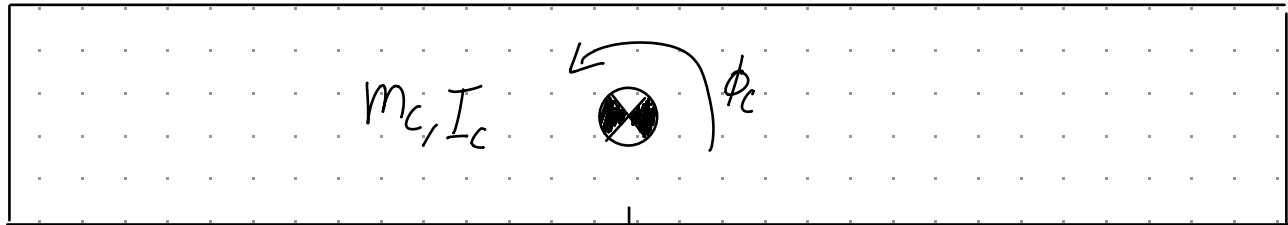
Half-Car Lumped-Parameter Model

Assumptions

- Small angles
- Rigid-Bodies

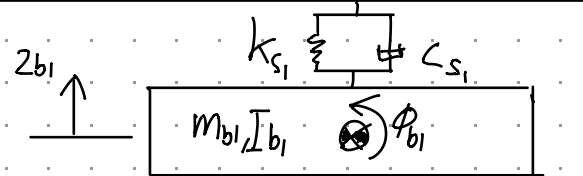
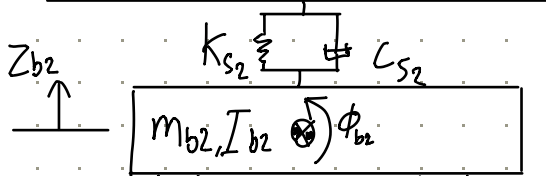


Car-
Body



z_c
↑

Secondary-
Suspension

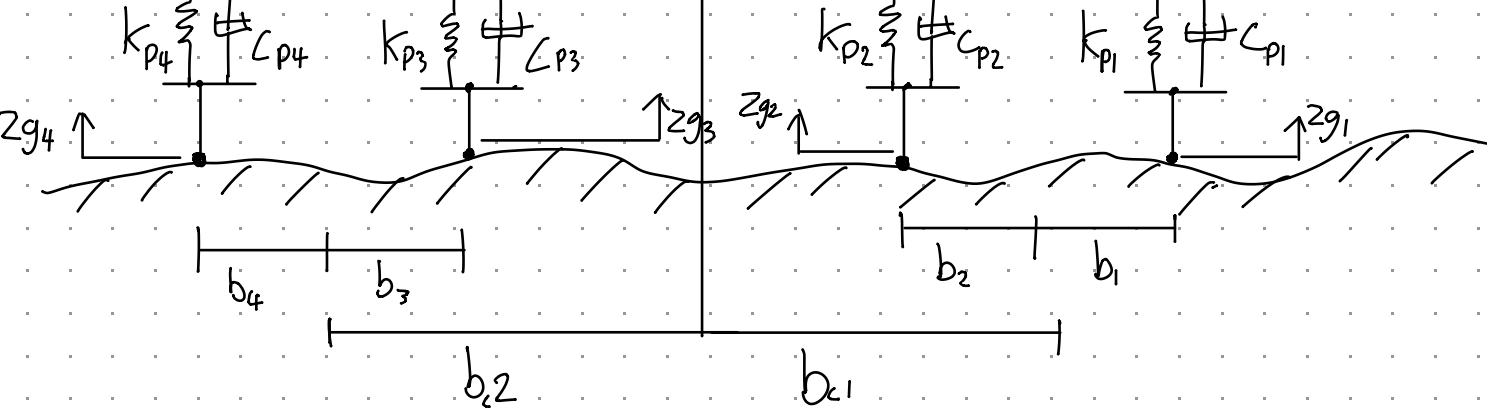


Bogies

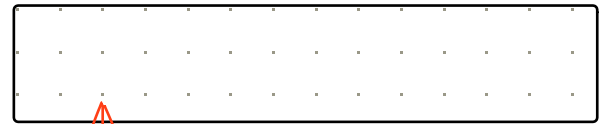
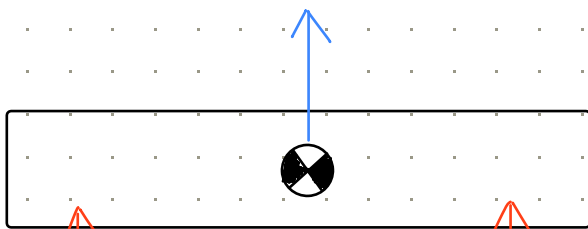
Primary-
Suspension

Wheels

Rail:



$$z_{b2} = (z_{b2} - b_4 \phi_{b2}) + (z_{b2} + b_3 \phi_{b2})$$



$$\begin{aligned} & -k_{p4}[(z_{b2} - b_4 \phi_{b2}) - z_{g4}] - c_{p4}[(\dot{z}_{b2} - b_4 \dot{\phi}_{b2}) - \dot{z}_{g4}] \\ & -k_{p3}[(z_{b2} + b_3 \phi_{b2}) - z_{g3}] - c_{p3}[(\dot{z}_{b2} + b_3 \dot{\phi}_{b2}) - \dot{z}_{g3}] \end{aligned}$$